Kitaev's ε-C^* rigidity beyond finite dimensions

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Abstract

Kitaev showed that a finite-dimensional ε-C^* algebra, which satisfies the C^* axioms up to errors of order ε, is $O(\varepsilon)$-isomorphic to a genuine C^* algebra, and that an almost-idempotent quantum channel on a finite-dimensional Hilbert space approximately factors through such an algebra. We study which of these statements survive in infinite dimensions. When the Hilbert space is infinite-dimensional but the almost-invariant algebra is finite-dimensional, as for codes with a finite-dimensional logical system, both results hold with the same universal constants, since the two Choi representations in Kitaev's proof can be replaced by Stinespring dilations. In the course of the proof we correct two points of the published argument, namely the diagonal of a block algebra with two or more blocks and the exact case of an idempotent channel. The results also extend to ℓ^∞-direct sums of finite-dimensional ε-C^* algebras, because Kitaev's constants are uniform. For a general infinite-dimensional ε-C^* algebra the question is open. We identify where Kitaev's proof uses finite dimension, show that there is no local obstruction, and point to degree-three bounded cohomology of non-amenable groups as the likely source of a counterexample. Both positive results are verified in Lean 4 with Mathlib, conditional on the cited results of Kitaev's paper and on standard facts about completely positive maps.

All equation, theorem and section numbers of the form (12.37), Theorem 2.3, §12 refer to Kitaev, *Almost-idempotent quantum channels and approximate C^*-algebras*, arXiv:2405.02434v2 [1].

1 The question, and the short answer

Kitaev's Theorem 2.3 [1] says that there are universal constants $\varepsilon_0, C_{\mathrm{K}}$ such that every *finite-dimensional* ε-C^* algebra $\mathcal{A}$ with $\varepsilon < \varepsilon_0$ admits a linear bijection $v\colon \mathcal{B} \to \mathcal{A}$ from a genuine C^* algebra $\mathcal{B}$ which is a $C_{\mathrm{K}}\varepsilon$-isomorphism: $v(X^\dagger) = v(X)^\dagger$, $\|v(I) - I\| \le C_{\mathrm{K}}\varepsilon$, $\|v(XY) - v(X)v(Y)\| \le C_{\mathrm{K}}\varepsilon\|X\|\|Y\|$ and $(1 - C_{\mathrm{K}}\varepsilon)\|X\| \le \|v(X)\| \le (1 + C_{\mathrm{K}}\varepsilon)\|X\|$. The question is whether this survives in infinite dimensions. "Infinite dimensions" can mean two different things, and the answers differ.

(a) **The physical Hilbert space $\mathcal{H}$ is infinite-dimensional, but the almost-invariant algebra $\mathcal{A} = \operatorname{Im}\widetilde{\Phi}$ is finite-dimensional.** This is the situation of bosonic or field-theoretic codes with a finite-dimensional logical system. Here the answer is *yes*: both the bijection v and Kitaev's UCP factorization $\Phi \approx \Delta\Upsilon$ (Theorem 12.3) hold, with the same universal constants and no assumption on $\dim\mathcal{H}$ (Theorem A). Kitaev states Theorem 12.3 only for $\dim\mathcal{H} < \infty$; the proof uses finite dimensionality of $\mathcal{H}$ only through two Choi representations, and both can be replaced by Stinespring dilations [4, 5]. Theorem A is machine-checked in Lean, conditional on the cited results of Kitaev's paper and three textbook facts about completely

positive maps (§5). The formalization exposed two points in the published argument that need repair, and both are fixed below.

(b) **The ε-C^* algebra $\mathcal{A}$ itself is infinite-dimensional.** If $\mathcal{A}$ is an ℓ^∞-direct sum of finite-dimensional pieces (infinitely many finite sectors) the answer is *yes*, because Kitaev's constants are uniform (Theorem B). We have verified this reduction in Lean 4 with Mathlib, together with its consequence for block-diagonal channels, with Kitaev's results as explicit hypotheses (§5). For a general infinite-dimensional $\mathcal{A}$ the question is *open*. We can neither prove nor refute it, and we found no claim either way in the literature (Kitaev [1]; Ranard, Walter and Witteveen [2], who use the finite-dimensional theorem; Alekseev, Farah and Thom [3], who remark that ε-C^* algebras are "subtly different" from Ulam stability [13, 14]). Section 4 explains where Kitaev's proof needs $\dim \mathcal{A} < \infty$. It also shows that no counterexample can be detected on finitely many elements, and identifies the most plausible place one could come from: degree-3 bounded cohomology of non-amenable groups.

2 Infinite-dimensional $\mathcal{H}$, finite-dimensional logical algebra

Throughout, $\Phi\colon \mathbf{B}(\mathcal{H}) \to \mathbf{B}(\mathcal{H})$ is UCP, $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \le \eta$, $\widetilde{\Phi} = \theta(2\Phi - 1)$ is defined by (12.2), $\mathcal{A} = \operatorname{Im}\widetilde{\Phi}$, and $X \star Y = \widetilde{\Phi}(XY)$.

Lemma 2.1. *If $\eta < 1/4$ then $\widetilde{\Phi} = \Phi\Psi$ for a bounded map Ψ commuting with Φ. Hence $\operatorname{Im}\widetilde{\Phi} \subseteq \operatorname{Im}\Phi$ and $\operatorname{rank}\widetilde{\Phi} \le \operatorname{rank}\Phi$.*

Proof. Let $N = 4(\Phi - \Phi^2) = 4\Phi(1 - \Phi)$, so $\|N\|_{\mathrm{cb}} \le 4\eta < 1$, and $(1 - x)^{-1/2} = \sum_{k\ge 0} c_k x^k$ with $c_0 = 1$. Then (12.2) reads

$$\widetilde{\Phi} = \tfrac{1}{2}\Big(1 + (2\Phi - 1)\sum_{k\ge 0} c_k N^k\Big) = \Phi + (\Phi - \tfrac{1}{2})\sum_{k\ge 1} c_k N^k = \Phi\Big[1 + 4(\Phi - \tfrac{1}{2})(1 - \Phi)\sum_{k\ge 1} c_k N^{k-1}\Big],$$

using $N^k = \Phi \cdot 4(1 - \Phi)N^{k-1}$. The bracket converges absolutely and commutes with Φ. □

Theorem A. *There are universal constants $\eta_0, C > 0$ with the following property. Let $\mathcal{H}$ be a nonzero Hilbert space of arbitrary dimension and $\Phi\colon \mathbf{B}(\mathcal{H}) \to \mathbf{B}(\mathcal{H})$ a UCP map with $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \le \eta \le \eta_0$ such that $\widetilde{\Phi}$ has finite rank (by Lemma 2.1 it suffices that Φ has finite rank). Then*

(i) there are a finite-dimensional C^* algebra $\mathcal{B}$ and an extended $C\eta$-isomorphism $v\colon \mathcal{B} \to \mathcal{A}$; in particular v is a linear bijection;

(ii) there are UCP maps $\Delta\colon \mathcal{B} \to \mathbf{B}(\mathcal{H})$ and $\Upsilon\colon \mathbf{B}(\mathcal{H}) \to \mathcal{B}$ with $\|\Delta\Upsilon - \Phi\|_{\mathrm{cb}} \le C\eta$ and $\|\Upsilon_n(\Delta_n(X)\Delta_n(Y)) - XY\| \le C\eta\|X\|\|Y\|$ for all n and $X, Y \in \mathbf{M}_n \otimes \mathcal{B}$; in particular $\|\Upsilon\Delta - 1_{\mathcal{B}}\|_{\mathrm{cb}} \le C\eta$.

Proof. (i) Kitaev proves Theorem 12.1 for arbitrary $\mathcal{H}$; the "Stinespring stack" of §12.1 is there for exactly this purpose. So $\mathcal{A}$, with the matrix norms inherited from $\mathbf{M}_n(\mathbf{B}(\mathcal{H}))$, is an extended $O(\eta)$-C^* algebra. It is finite-dimensional by hypothesis. Theorem 10.5 is a statement about abstract finite-dimensional extended ε-C^* algebras and makes no reference to $\mathcal{H}$, so it gives $\mathcal{B}$ and v.

(ii) Kitaev's proof of Theorem 12.3 consists of the outline (12.25)–(12.27), the construction of Δ' and Δ, the properties (12.33)–(12.36), Lemma 12.4, and the construction of Υ' and Υ. The outline, Δ, and (12.33)–(12.36) use only the diagonal of the finite-dimensional algebra $\mathcal{B}$, the bounds of Theorem 12.1 (including (12.6) applied to Φ_n), and normalization by $\Delta'(I)^{-1/2}$. None of these uses $\dim \mathcal{H} < \infty$.

One correction is needed in this part. Kitaev represents the diagonal of each block $\mathbf{B}(\mathcal{L}_j)$ by a unitary 1-design $D_j = \sum_s p_{js} U_{js}^\dagger \otimes U_{js}$ satisfying (12.29), and takes as the diagonal of $\mathcal{B}$ the product $D = \sum_s p_s U_s^\dagger \otimes U_s$, with $p_s = \prod_j p_{js_j}$ and $U_s = \bigoplus_j U_{js_j}$. For two or more blocks this D does not satisfy $XD = DX$. For $\mathcal{B} = \mathbb{C} \oplus \mathbb{C}$ with trivial designs, $D = 1 \otimes 1$, and $X = (1, 0)$ gives $XD \neq DX$. The cross-block terms survive because they carry the first moments $\sum_s p_{js} U_{js}$, which need not vanish. The identity $XD = DX$ is what makes Δ' completely positive. It is restored by attaching an independent uniform sign to each block, $U_{s,\sigma} = \bigoplus_j \sigma_j U_{js_j}$ with $\sigma \in \{\pm 1\}^m$ uniformly distributed. Every cross-block term then averages to zero, each diagonal block reproduces (12.29), and nothing else in the argument changes.

Finite dimensionality of $\mathcal{H}$ enters only through two Choi representations, which we replace as follows.

(a) The representation (12.37) of Δ. Write $\mathcal{B} = \bigoplus_{j=1}^m \mathbf{B}(\mathcal{L}_j)$, $\dim \mathcal{L}_j < \infty$. By Stinespring's theorem [4], [5, Thm. 4.1], $\Delta(X) = W^\dagger \rho(X) W$ with ρ a unital $*$-representation of $\mathcal{B}$ on some $\mathcal{K}$ and W an isometry. Every unital representation of $\bigoplus_j \mathbf{B}(\mathcal{L}_j)$ is unitarily equivalent to $X \mapsto \bigoplus_j X_j \otimes 1_{\mathcal{E}_j}$ on $\bigoplus_j \mathcal{L}_j \otimes \mathcal{E}_j$, for Hilbert spaces $\mathcal{E}_j$ of arbitrary dimension (possibly 0). Writing $W = (W_1, \dots, W_m)$ gives exactly (12.37), with $\sum_j W_j^\dagger W_j = W^\dagger W = 1_{\mathcal{H}}$.

(b) Lemma 12.4. The map $a \otimes b \mapsto (a \otimes 1) W_j W_j^\dagger (b \otimes 1)$ is linear on the algebraic tensor product $\mathbf{B}(\mathcal{L}_j) \otimes \mathbf{B}(\mathcal{L}_j)$. Applying it to (12.29) shows that R_j in (12.39) commutes with $\mathbf{B}(\mathcal{L}_j) \otimes 1_{\mathcal{E}_j}$. The commutant of $\mathbf{B}(\mathcal{L}_j) \otimes 1$ in $\mathbf{B}(\mathcal{L}_j \otimes \mathcal{E}_j)$ is $1 \otimes \mathbf{B}(\mathcal{E}_j)$ whatever the dimension of $\mathcal{E}_j$, so $R_j = 1 \otimes C_j$. The bounds $1 - O(\eta) \le \|C_j\| \le 1$ use only (12.33) and (12.35). The one change is that, when $\dim \mathcal{E}_j = \infty$, $\|C_j\|$ need not be attained. For $\eta > 0$ we choose a unit vector ξ_j with $\|C_j \xi_j\| \ge \|C_j\| - \eta \ge 1 - O(\eta)$, which changes only the constant. For $\eta = 0$ this choice is not available, while the theorem then asks for an exact factorization; this case is treated separately in (d).

(c) The Choi representation $\Phi(X) = V^\dagger (X \otimes 1_{\mathcal{F}}) V$ *[6].* This exists only for normal Φ. Instead take a Stinespring dilation $\Phi(X) = V^\dagger \pi(X) V$, with $\pi \colon \mathbf{B}(\mathcal{H}) \to \mathbf{B}(\mathcal{K}')$ a unital $*$-representation and V an isometry, and define

$$L_j = \sum_s p_{js}\, \pi\big(\Delta(U_{js}^\dagger)\big)\, V\, W_j^\dagger (U_{js} \otimes \xi_j) \colon \mathcal{L}_j \to \mathcal{K}', \qquad \Upsilon'_j(X) = L_j^\dagger\, \pi\big(\Phi(X)\big)\, L_j .$$

Each Υ'_j is completely positive and $\|L_j\| \le 1$. Since $\pi(\Delta(U^\dagger))^\dagger = \pi(\Delta(U))$ and

$$V^\dagger \pi(Z_1) \pi(Z_2) \pi(Z_3) V = \Phi(Z_1 Z_2 Z_3),$$

we get

$$\Upsilon'_j(\Delta(Y)) = \sum_{r,s} p_{jr} p_{js}\, (U_{jr}^\dagger \otimes \xi_j^\dagger)\, W_j\, \Phi\big(\Delta(U_{jr})\, \Phi(\Delta(Y))\, \Delta(U_{js}^\dagger)\big)\, W_j^\dagger (U_{js} \otimes \xi_j).$$

This is the first line of Kitaev's computation of $\Upsilon'_j(\Delta(Y))$. The subsequent lines use (12.34), (12.36), (12.37), (12.39) and Lemma 12.4, all of which hold by (a) and (b). At matrix level n, the dilation $(1_n \otimes V,\, 1_n \otimes \pi)$ of Φ_n gives $(\Upsilon'_j)_n(X) = (1 \otimes L_j)^\dagger (1 \otimes \pi)(\Phi_n(X))(1 \otimes L_j)$, so the same computation applies. Likewise $\Upsilon'_j \Phi - \Upsilon'_j = L_j^\dagger \pi\big((\Phi^2 - \Phi)(\cdot)\big) L_j$ has cb-norm at most η. The chain $\Upsilon' \approx \Upsilon' \Phi \approx \Upsilon' \widetilde{\Phi} = \Upsilon' \widetilde{\Delta} \widetilde{\Upsilon} \approx \Upsilon' \Delta \widetilde{\Upsilon} \approx \widetilde{\Upsilon}$ and the normalization $\Upsilon = \Upsilon'(1)^{-1/2} \Upsilon'(\cdot) \Upsilon'(1)^{-1/2}$ are dimension-free.

(d) The case $\eta = 0$. Then $\Phi^2 = \Phi$, so $\widetilde{\Phi} = \Phi$, and Theorem 10.5 gives $\delta = 0$. So v is a unital complete isometry with $\Phi(v(X) v(Y)) = v(XY)$, hence $\Delta' = v$ and $\Delta = v$. Take $\Upsilon = \widetilde{\Upsilon} = v^{-1} \Phi$. It is unital, and it is completely positive. If $X \ge 0$ in $\mathbf{M}_n(\mathbf{B}(\mathcal{H}))$, then $Y = \Upsilon_n(X)$ is self-adjoint

and $v_n(Y) = \Phi_n(X) \geq 0$. With $t = \|Y\| = \|v_n(Y)\|$ we get $\|t - Y\| = \|t - v_n(Y)\| \leq t$, and a self-adjoint element with $\|t - Y\| \leq t$ is positive. Finally $\Delta\Upsilon = \Phi$ and $\Upsilon_n(\Delta_n(X)\Delta_n(Y)) = v_n^{-1}\big(\Phi_n(v_n(X)v_n(Y))\big) = XY$ exactly. □

Remark 2.2. A simpler choice of ξ_j also works, although it is not part of the Lean proof. The computation in (c) gives $\Upsilon'\Delta \approx D$ rather than ≈ 1, where $D = (\|C_j\xi_j\|^2)_j$, so the chain yields $\Upsilon' = D\widetilde{\Upsilon} + O(\eta)$. Since D is central in $\mathcal{B}$, $\Upsilon'(1)^{-1/2} = D^{-1/2} + O(\eta)$, and the normalization divides D out. So any ξ_j with $\|C_j\xi_j\| \geq 1/2$ suffices, uniformly in $\eta \geq 0$, and no approximate maximizer is needed.

Remark 2.3. The rank hypothesis is about the spectrum of Φ near 1, not about $\mathcal{H}$. Since $\|\Phi^2 - \Phi\|_{\mathrm{cb}} < 1/4$, $\sigma(\Phi)$ splits into a part near 0 and a part near 1, and $\widetilde{\Phi}$ is the Riesz projection onto the latter. So $\widetilde{\Phi}$ has finite rank exactly when that part consists of finitely many eigenvalues of finite algebraic multiplicity. This holds, for instance, whenever Φ is a compact operator on the Banach space $\mathbf{B}(\mathcal{H})$. The case $\Phi = 1$ (where $\mathcal{A} = \mathbf{B}(\mathcal{H})$ is already a C^* algebra) shows that the hypothesis is not automatic.

Corollary 2.4 (codes through a finite-dimensional system)**.** *Let $\mathcal{C} \subseteq \mathcal{H}$ be a finite-dimensional subspace and T a channel on trace-class operators whose outputs are supported in $\mathcal{C}$, for example noise followed by a recovery map into a code space. If $\Phi = T^*$ satisfies $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \leq \eta \leq \eta_0$, then (i) and (ii) of Theorem A hold.*

Proof. Since $T(\rho) = P_{\mathcal{C}}T(\rho)P_{\mathcal{C}}$, we have $\Phi(X) = \Phi(P_{\mathcal{C}}XP_{\mathcal{C}})$, so $\operatorname{rank}\Phi \leq (\dim\mathcal{C})^2$. Apply Lemma 2.1 and Theorem A. □

3 Infinitely many finite sectors

Proposition 3.1. *Let $I \neq \emptyset$. For each $i \in I$ let $\mathcal{A}_i$ be an ε-C^* algebra, $\mathcal{B}_i$ a C^* algebra, and $v_i\colon \mathcal{B}_i \to \mathcal{A}_i$ a δ-isomorphism, with the same ε and the same $\delta < 1$ for all i. Let $\mathcal{A} = \bigoplus_i^\infty \mathcal{A}_i$ and $\mathcal{B} = \bigoplus_i^\infty \mathcal{B}_i$ be the bounded families with supremum norm and coordinatewise operations. Then $\mathcal{A}$ is an ε-C^* algebra, $\mathcal{B}$ is a C^* algebra, and $v = (v_i)_i\colon \mathcal{B} \to \mathcal{A}$ is a δ-isomorphism. The same holds for extended structures.*

Proof. Every axiom (2.1)–(2.6) is an inequality between quantities computed coordinatewise. It holds in each coordinate with the same ε, so taking suprema gives it in $\mathcal{A}$. For instance $(1-\varepsilon)\|a_i\|^2 \leq \|a_i^\dagger a_i\| \leq \|a^\dagger a\|$ for every i, hence $(1-\varepsilon)\|a\|^2 \leq \|a^\dagger a\|$. The bound $|\|I\| - 1| \leq \varepsilon$ is where $I \neq \emptyset$ is needed: $\|I\| = \sup_i \|I_i\| \in [1-\varepsilon, 1+\varepsilon]$. Next, $\|v(b)\| = \sup_i \|v_i(b_i)\|$ lies in $[(1-\delta)\|b\|, (1+\delta)\|b\|]$, so v is well defined and bounded. The inverse $(v_i^{-1})_i$ has norm at most $(1-\delta)^{-1}$, so v is bijective. The unit and multiplicativity defects are coordinatewise at most δ and $\delta\|b_i\|\|c_i\| \leq \delta\|b\|\|c\|$. For extended structures use $\mathbf{M}_n(\bigoplus^\infty \mathcal{A}_i) = \bigoplus^\infty \mathbf{M}_n(\mathcal{A}_i)$ isometrically. □

Theorem B. *Let $\varepsilon_0, C_{\mathrm{K}}$ be the constants of Kitaev's Theorem 2.3 (or 10.5). Every ℓ^∞-direct sum of finite-dimensional (extended) ε-C^* algebras with $\varepsilon < \varepsilon_0$ and $C_{\mathrm{K}}\varepsilon < 1$ is (extended) $C_{\mathrm{K}}\varepsilon$-isomorphic to a C^* algebra. Consequently, if $\mathcal{H} = \bigoplus_k \mathcal{H}_k$ and $\Phi(X) = \bigoplus_k \Phi_k(P_kXP_k)$, where each Φ_k is UCP on $\mathbf{B}(\mathcal{H}_k)$ with $\|\Phi_k^2 - \Phi_k\|_{\mathrm{cb}} \leq \eta \leq \eta_0$ and $\widetilde{\Phi}_k$ of finite rank, and $C\eta < 1$, with η_0, C the constants of Theorem A, then $\|\Phi^2 - \Phi\|_{\mathrm{cb}} \leq \eta$, and (i) and (ii) of Theorem A hold with the infinite-dimensional algebra $\mathcal{B} = \bigoplus_k^\infty \mathcal{B}_k$.*

Proof. The first claim follows from Proposition 3.1 by choosing, for each i, Kitaev's $\mathcal{B}_i$ and v_i. For the channel statement, $\Phi^m(X) = \bigoplus_k \Phi_k^m(P_kXP_k)$ for $m \geq 1$. The compression $X \mapsto P_kXP_k$

is completely positive and completely contractive, and the norm of a matrix of block-diagonal operators is the supremum over k of the norms of its k-th diagonal blocks, so $\|\Phi^2-\Phi\|_{\mathrm{cb}} \le \eta$. By the proof of Lemma 2.1, $\widetilde{\Phi} = \Phi\Psi$ is a norm-convergent series in Φ without constant term. Hence $\widetilde{\Phi}(X) = \bigoplus_k \widetilde{\Phi}_k(P_k X P_k)$. Since $\widetilde{\Phi}_k$ is the identity on $\operatorname{Im}\widetilde{\Phi}_k$, this gives $\mathcal{A} = \bigoplus_k^{\infty} \operatorname{Im}\widetilde{\Phi}_k$ with coordinatewise Choi–Effros product [7]. Theorem A gives extended $C\eta$-isomorphisms $v_k\colon \mathcal{B}_k \to \operatorname{Im}\widetilde{\Phi}_k$, and $C\eta < 1$ lets Proposition 3.1 assemble them. Put $\Delta = \bigoplus_k \Delta_k$ and $\Upsilon(X) = (\Upsilon_k(P_k X P_k))_k$, both UCP. All the required bounds are suprema of the bounds of Theorem A for the Φ_k. □

Remark 3.2. The same argument shows that ultraproducts $\prod_{\mathcal{U}} \mathcal{A}_i$ of finite-dimensional ε-C^* algebras are $C_{\mathrm{K}}\varepsilon$-isomorphic to $\prod_{\mathcal{U}} \mathcal{B}_i$: the maps v_i and v_i^{-1} are uniformly bounded, so $\prod_{\mathcal{U}} v_i$ is a well-defined bijection, and the defects pass to the limit. Thus the finite-dimensional theorem with *uniform* constants already covers every infinite-dimensional algebra that is assembled from finite-dimensional ones in a uniformly bounded way.

4 Why the general infinite-dimensional case is open

4.1 The three places where $\dim \mathcal{A} < \infty$ is load-bearing

1. *The first projection (Lemma 6.1).* The induction starts from a nontrivial approximate projection, found as a fixed point of inversion on the approximate unitary group via the Lefschetz–Hopf theorem [8]. That group is a compact manifold only when $\dim \mathcal{A} < \infty$. In infinite dimensions the conclusion of the lemma is false even for genuine C^* algebras: $C[0,1]$ and $C_r^*(F_2)$ have no nontrivial projections. So no argument can supply the first step; a proof would have to start from something else, for example an approximately commutative subalgebra close to some $C(K)$, which would need an approximate Gelfand theory that does not exist yet.

2. *Termination (§9).* Merging and extension stop because $\dim \mathcal{A}$ is finite. Each step re-runs error reduction, which moves the whole map by $O(\varepsilon)$, so an infinite sequence of steps need not converge.

3. *Error reduction (Lemma 8.2).* The Newton step uses an exact contractive diagonal $D \in \mathcal{B}\hat{\otimes}\mathcal{B}$. A C^* algebra with a bounded diagonal is finite-dimensional (contractible). For infinite-dimensional $\mathcal{B}$ the natural replacement is Johnson's bounded approximate diagonal [9, 10], which exists exactly when $\mathcal{B}$ is nuclear [11, 12], and which needs weak* limits in the target. For non-nuclear $\mathcal{B}$ (such as $\mathbf{B}(\ell^2)$ or $C_r^*(F_2)$) there is no diagonal of any kind, and Ulam-type stability is known to fail in neighbouring problems [13, 14]. Note that Corollary 8.3 does allow an infinite-dimensional *target*; what is missing is the construction of $\mathcal{B}$.

4.2 No local obstruction

Proposition 4.1. *For every $k \in \mathbb{N}$ and $\delta > 0$ there is $\varepsilon(k,\delta) > 0$ such that the following holds. For every $\varepsilon(k,\delta)$-C^* algebra $\mathcal{A}$ and every $x_1, \dots, x_k$ in its unit ball, there are a C^* algebra $\mathcal{B}$ and $y_1, \dots, y_k \in \mathcal{B}$ with $|\|w(x)\| - \|w(y)\|| \le \delta$ for every $*$-word w of length at most k, evaluated with a fixed bracketing.*

Proof. If not, there are k, δ and ε_n-C^* algebras $\mathcal{A}_n$ with $\varepsilon_n \to 0$ and tuples $x^{(n)}$ for which every pair $(\mathcal{B}, y)$ fails. Fix a free ultrafilter $\mathcal{U}$. The ultraproduct $\prod_{\mathcal{U}} \mathcal{A}_n$, with coordinatewise operations, is well defined because the products are uniformly bounded. It satisfies (2.1)–(2.6) with $\varepsilon = \lim_{\mathcal{U}} \varepsilon_n = 0$,

so it is a C^* algebra. Take $\mathcal{B}$ to be this ultraproduct and $y_i = [x_i^{(n)}]$. Then $\|w(y)\| = \lim_{\mathcal{U}} \|w(x^{(n)})\|$ for each of the finitely many words, so for $\mathcal{U}$-most n all of them are within δ, a contradiction. □

So any counterexample must be global: it can only be witnessed by configurations of unbounded size or complexity. By Theorem B and the remark after it, it also cannot be assembled from finite-dimensional pieces in a bounded way.

4.3 Where a counterexample would have to come from

The phenomenon "locally small error, globally far from exact" is familiar from approximate group homomorphisms. Its degree-two form is the following.

Proposition 4.2 (warm-up: quasimorphisms)**.** *Let $f\colon \Gamma \to \mathbb{R}$ be a homogeneous quasimorphism with defect D that is not a homomorphism (such f exist for $\Gamma = F_2$ [15]), and let $0 < t < 2\pi/D$. Then $\chi_t = e^{itf}$ satisfies $|\chi_t(gh) - \chi_t(g)\chi_t(h)| \le tD$. Yet for every Hilbert space $\mathcal{K}$ there is no homomorphism $\pi\colon \Gamma \to U(\mathcal{K})$ with $\sup_g \|\pi(g) - \chi_t(g)1\| < \sqrt{3}$.*

Proof. Suppose π exists. Let $\lambda = \chi_t(g)e^{i\alpha} \in \sigma(\pi(g))$. Then $\lambda^n \in \sigma(\pi(g^n))$ and $\chi_t(g^n) = \chi_t(g)^n$ by homogeneity, so $|e^{in\alpha} - 1| < \sqrt{3}$ for all n. This forces $\alpha \in 2\pi\mathbb{Z}$: otherwise some multiple of α lies in $[2\pi/3, 4\pi/3]$ mod 2π. Hence $\pi(g) = \chi_t(g)1$ for all g, so χ_t is multiplicative. Then $t(f(gh) - f(g) - f(h)) \in 2\pi\mathbb{Z}$, with absolute value at most $tD < 2\pi$, so f is a homomorphism, a contradiction. □

Associativity is a degree-three condition, so the analogue lives in H^3_b. Let Γ be a group with a bounded real 3-cocycle β whose class is nonzero in $H^3_b(\Gamma;\mathbb{R})$ but zero in $H^3(\Gamma;\mathbb{R})$. For closed surface groups $\Gamma = \pi_1\Sigma_g$ with $g \ge 2$, $H^3(\Gamma;\mathbb{R}) = 0$ while $H^3_b(\Gamma;\mathbb{R})$ is infinite-dimensional [16]. Write $\beta = dc$ with an unbounded 2-cochain c, and twist the group algebra by

$$e_g \star_t e_h = e^{itc(g,h)} e_{gh}, \qquad \text{so that} \qquad (e_g \star_t e_h) \star_t e_k = e^{-it\beta(g,h,k)}\, e_g \star_t (e_h \star_t e_k).$$

On basis vectors the associator is at most $t\|\beta\|_\infty$, which is arbitrarily small. One step of the analogue of Proposition 4.2 is elementary.

Proposition 4.3 (graded rigidity)**.** *If $t\|\beta\|_\infty < \pi$, there is no $U(1)$-valued 2-cocycle σ (that is, no associative twist) with $\sup_{g,h} |e^{itc(g,h)} - \sigma(g,h)| < 2\sin(\pi/8) \approx 0.77$.*

Proof. Write $e^{itc}\sigma^{-1} = e^{i\gamma}$ with $|\gamma| < \pi/4$ and $\sigma = e^{i\tau}$. Then $tc = \tau + \gamma + 2\pi m$ with m integer-valued, and $d\tau$ is $2\pi\mathbb{Z}$-valued because σ is a cocycle. Applying d shows that $t\beta - d\gamma$ is $2\pi\mathbb{Z}$-valued. But $|t\beta - d\gamma| < \pi + 4 \cdot \pi/4 = 2\pi$, so $t\beta = d\gamma$. Hence $\beta = d(\gamma/t)$ with γ/t bounded, i.e. $[\beta] = 0$ in H^3_b, a contradiction. □

Two steps separate this from a counterexample to the infinite-dimensional theorem, and we could not complete either.

1. *A C^*-type norm.* In $\ell^1(\Gamma)$ the basis-vector bounds extend to all elements, but ℓ^1 violates the C^* identity badly. For a norm satisfying (2.5) one needs the phases $e^{-it\beta} - 1$ to be small as *multipliers* on the relevant tensor products, which is much stronger than smallness in sup norm. Physically, this is Jackiw's non-associativity of magnetic translations [17] in the presence of monopole density: in hyperbolic space, uniform charge density gives a uniformly bounded charge in every geodesic tetrahedron, so the associator is uniformly small, and the obstruction is a bounded volume-type class.

2. *Ungraded rigidity.* Proposition 4.3 excludes nearby *graded* associative twists. A counterexample needs to exclude every C^* algebra, i.e. the analogue of the spectral step in Proposition 4.2. For non-nuclear algebras there is no spectral theory of the relevant elements to borrow.

4.4 What we expect

Our guess is that the infinite-dimensional statement is true for ε-C^* algebras that are "nuclear-like", where approximate diagonals are available. We suspect that it fails in general, with the counterexample living where bounded cohomology of non-amenable groups lives. This is a guess, not a result.

5 What is verified in Lean

Both theorems are machine-checked in Lean 4 [18] with Mathlib [19] (toolchain `v4.35.0-rc4`). Theorem A is one certificate and Theorem B is two. Each certificate states its result as a single proposition `Claim`, proves it as `main : Claim`, and passes an audit. The audit scans for `sorry`, `admit` and new axioms, and checks that `main` proves exactly `Claim`. It confirms with `#print axioms` that only Lean's standard axioms (`propext`, `Classical.choice`, `Quot.sound`) are used, and replays every module in an independent kernel checker. Results that are not formalized enter only as named hypotheses of `Claim`, never as axioms. The repository of this paper contains the Lean sources (`lean_sources.zip`) and the certificate `lean_certificate.pdf`, which gives the formal statements, the assumptions and a step-by-step map from the text to the Lean lemmas.

- *Theorem A*, conditional on six named results. Three are textbook facts about completely positive maps: $\|\varphi\|_{\mathrm{cb}} \le \|\varphi(1)\|$ and $\varphi(x^\dagger) = \varphi(x)^\dagger$ for CP φ; Stinespring's theorem; and the existence of unitary 1-designs. The other three are Kitaev's Theorem 12.1, equation (12.6) and Theorem 10.5, each with its $O(\cdot)$ made explicit as a constant. Everything else is proved: the construction of Δ, the properties (12.34)–(12.36) at every matrix level, Lemma 12.4, the choice of ξ_j, the construction of Υ, both normalizations, and the case $\eta = 0$. The constants come out explicit: $\eta_0 = \min\bigl(\eta_1, \eta_2, \varepsilon_3/(c_1+1), 1/(2\cdot 10^5\Lambda)\bigr)$ and $C = 10^6\Lambda$, with $\Lambda = C_3c_1 + c_2 + 1$, where (c_1, η_1), (c_2, η_2), (C_3, ε_3) are the constants of the three cited results. A non-vacuity check shows that the hypotheses hold on $\ell^2(\mathbb{N})$ for $\Phi(X) = \langle e_0, Xe_0\rangle 1$, which is UCP, idempotent and of rank one.

- *Theorem B* in full, together with Proposition 3.1. The first sentence is certified for ε-C^* algebras, conditional on Kitaev's Theorem 2.3, and for extended ones, conditional on Theorem 10.5. Extended algebras are realized inside $\mathbf{B}(\mathcal{H}_k)$, and their sum as block-diagonal operators on $\bigoplus_k \mathcal{H}_k$. The second sentence is certified under the six hypotheses of Theorem A. The formalization showed that its condition $C\eta < 1$ is needed: Proposition 3.1 requires $\delta < 1$, and the constants of Theorem A allow $C\eta_0$ to be as large as 5. Non-vacuity checks show that the hypotheses hold with infinitely many sectors; that at $\varepsilon = 0$ the transcribed Definitions 2.1–2.2 rebuild Mathlib's `CStarAlgebra`; and that the pinching $X \mapsto \bigoplus_k P_k X P_k$ on $\ell^2(\mathbb{N})$ satisfies the hypotheses of the second sentence while its $\widetilde{\Phi}$ has infinite rank, so that sentence is not a case of Theorem A.

- Also formalized, with only the standard axioms but outside the certificate audit: the inverse of a δ-isomorphism is a $\delta/(1-\delta)^3$-isomorphism (Kitaev's "$\delta + O(\delta^2)$" with an explicit constant), and Proposition 4.3 in full.

Not formalized: Lemma 2.1, Corollary 2.4, the remark following Theorem A, the remark following Theorem B, and Section 4 apart from Proposition 4.3. Kitaev's own proofs of Theorems 2.3 and 10.5 (Lefschetz–Hopf, H-spaces, the implicit function theorem on the approximate unitary group) are far beyond what Mathlib supports, so they enter only as hypotheses.

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